My Notes

ASA Firewall Configuration

STEP 1: Review the Network Diagram before adding the Firewall

1) Office (Network) ---> 2 PCs + Access-SW
Subnet: 192.168.1.0 /24
PC1 IP address: 192.168.1.10
PC2 IP address: 192.168.1.20
Public Server IP address: 198.51.100.2

2) Edge Router

3) ISP Router + Public server simulating the Internet

STEP 2: Review the Edge Router Current Configuration

G0/0 (Interface facing the ISP Router) --> 203.0.113.2
G0/1 (Interface facing the Internal Network) ---> 192.168.1.1

Routes in the RTR routing table:
Local: 192.168.1.1 /32
Local: 203.0.113.2 /32

Connected: 192.168.1.0 /24
Connected: 203.0.113.0 /30

Please remember the following:
Connected Route (C): Represents a network directly connected to a router's interface.
Local Route (L): Represents the router's own interface IP address.

Static Default route: 0.0.0.0 0.0.0.0 203.0.113.1
Note: 203.0.113.1 is the next-hop address which is the ISP router

NAT Configuration
G0/0 = **ip nat outside**
G0/1 = **ip nat isnside**

PAT Configuration
ip nat inside source list 1 interface g0/0 overload

Explanation

ip nat inside source list 1 = translate the source IP addresses of the internal traffic and use ACL 1 to identify which internal IP addresses can be translated

interface g0/0 overload = use G0/0 IP address as the translated public IP and allow multiple devices to share the same IP using different port numbers

ACL Configuration

access-list 1 permit 192.168.1.0 0.0.0.255

STEP 3: Test Connectivity & Save the configuration files

Test Connectivity between PC 1 & PC 2 & the Public Server

2. Save the configuration files on the EDGE-RTR & Access-SW

STEP 4: Modify the Network Structure

1) Add the ASA Firewall between the EDGE-RTR and the Access-SW

Note: The Firewall model = ASA 5505 Firewall

Connect EDGE-RTR G0/0 with ASA Et0/0

Connect Access-SW G0/1 with ASA Et0/1

2) Remove the PAT Rule, ACL, then the NAT roles on each interface in the Edge Router

3) Assign new IP address for the Edge Router G0/1 interface

Edge Router G0/1 new IP address: 209.165.200.225 /29

4) Add a Static route on the ISP router pointing towards the Edge router so it can be able to send back replies back to the internal network

STEP 5: ASA Firewall Configuration

1) Enable the outside & inside Ethernet ports and assign each port to a separate VLAN

2) Give each VLAN a logical name, a security level & an IP address

VLAN 1 (Et0/1):

Name = inside

IP address = 192.168.1.1 255.255.255.0

VLAN 2 (Et0/0):

Name = outside

IP address = 209.165.200.226 /29

3) Disable the DHCP daemon on the inside port so ASA doesn't hand out addresses

4) Configure a default route on the ASA Firewall to send traffic that is not in its own network to the Edge Router (209.165.200.225)

5) Setup PAT on the ASA so all the PCs could share ASA outside address to access the Internet or outside Network

Explanation of the PAT configuration

object network Inside-Network

Creates a named object

subnet 192.168.1.0 255.255.255.0

Assign a network to the object

nat (inside,outside) dynamic interface

Translate traffic going from the inside to the outside and use different port numbers

6) Enable ICMP Inspection to allow the ASA to remember outgoing ping requests and let their replies come back in.

ICMP Inspection Explanation

class-map inspection_default

Creates a group for traffic

match default-inspection-traffic

Selects the traffic to inspect

policy-map global_policy

Creates a policy,

class inspection_default

Applies the next rule to that traffic

inspect icmp

Allows the ASA to track outgoing ping requests and let their replies come back in

service-policy global_policy global

Applies the policy to traffic globally